Scoring: use the criteria for Method .

Before its first collision with a plate, the ball has kinetic energy and speed

Just after this collision, the charge on the ball is

Its kinetic energy and speed are then

The electric force accelerates it toward the other plate. Just before its second collision, its kinetic energy is

Using gives

Its speed just before the second collision is

Its speed and kinetic energy just after the second collision are

From ,

Its kinetic energy just before the third collision with a plate is

From ,

Its speed just before the third collision with a plate is

Its speed and kinetic energy just after the third collision are